

On Representation Compression

What we know, is that [MBE compression] has causal effect on SCAN dataset (compositional generalization), when applied to Qwen model series. However, same is not true on other datasets, or pre-training, indicating that [MBE compression] is not the right causal lever. (And that they NYU folks are basically talking BS)

Perhaps we should not target "compression", but instead, we should target "composition". A useful analogy is "oligami". If we view "wrinkle" on the paper as an increase in entropy, then an "oligami" has minimal "wrinkles" locally but plenty of "wrinkle" globally, this creates an (elegant) structure. In comparison, a wad of paper (废纸团) is both locally and globally full of wrinkles, this means it does not have any elegant structure. Another object worth considering is a piece of paper, no wrinkles locally or globally, it has zero structure. [MBE compression is pointing towards piece of paper, what is missing, is likely the "global MBE incrementing force", which we can obtain via "accumulated trajectory level MBE maximization" in a similar design to "mutual information maximization"]

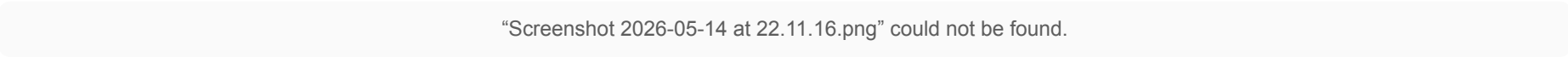


This is under the hypothesis that "compositional structure" don't discriminate against continuous representations.

[Winoground: VLM -> counterfactual, compositional generalization]
[The Illusion of Thinking] Used toy case math & puzzle
[VLA]

[MBE on SCAN] We observe compressing MBE has immense benefit on SCAN dataset, it avoids token level pattern matching, but exposes error on semantic level pattern matching (*a.k.a. it fail to count how many turns are required for making a circle*)

In [Vision Transformers Need More Than Registers] a smoothing is applied to the frequency space, although it's not clear what "Gaussian weight duplicated to all patches" mean here ... this points us to the direction of "low-pass filter / low-frequency filter" angle



A direct analogy on low rank matrix representation is to take a down projection matrix W_{down} , then set the up projection matrix as its pseudo inverse [requires: full rank down projection matrix]

$$W_{up} = (W_{down}^T W_{down})^{-1} W_{down}^T$$

If we have orthogonal columns for W_{down} , above equation simplifies into

$$W_{up} = W_{down}^T$$

(we can use Newton-Schulz Iteration to orthogonalize W_{down} to make above simple formulation valid, our learned "low rank representation" $\hat{R}$ would be

$$\hat{R} = R \times W_{down} \times W_{up}$$

here W_{down} is fully learnable and we can similarly obtain "stability score" on each tokens.

[This direction is directly aligned with the MBE reduction perspective, except that we are doing "reconstruction" based learning, so there are zero competing objectives induced what soever, in comparison, introducing explicit MBE regularization creates a competing objectives]

[Sinkhorn Knopp transformation to convert representation matrix into a doubly random matrix?]
This way we can probably truncate "not important dimensions" easier?

[Relative "compression ratio"]
Instead of just $r(R)$, try $\frac{r(R)}{r(E)}$ where E is the embedding matrix of a LLM --- but this is also susceptible to representation collapse, the real question is whether there exists a variants, which can't induce representation collapse and doesn't require "low-rank reconstruction"?

[Orthogonalization with Newton Schulz]
Given a matrix G , Singular Value Decomposition (SVD):

$$G = U \Sigma V^T$$

its "closest" orthogonal matrix is one that assigns same "weight" to all its eigen direction, that is

$$Q = UV^T$$

Newton-Schulz iteration approximates Q effectively via

$$X_{k+1} = 1.5X_k - 0.5X_kX_k^TX_k$$

Since $X_1 = G$, we have

$$X_1 = 1.5U\Sigma V^T - 0.5U\Sigma^3V^T = U(1.5\Sigma - 0.5\Sigma^3)V^T$$

to show the recursion converge to Q , we need to prove for sequence $\{a_n\}$ satisfying

$$a_{n+1} = 1.5a_n - 0.5a_n^3$$

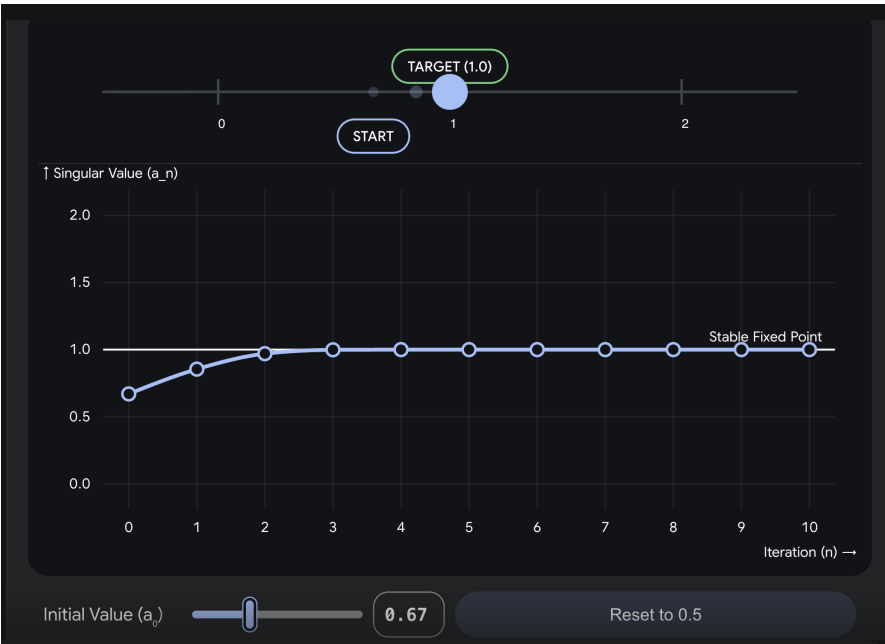
we have $\lim_{n \rightarrow \infty} a_n = 1$. We look for fixed point of the recursion, that is, when

$$a = 1.5a - 0.5a^3 \implies a(1 - a^2) = 0 \implies a = 0, 1, -1$$

we have 3 fixed points, consider derivative of $f(a) = 1.5a - 0.5a^3$, we know

$$f'(a) = 1.5 - 1.5a^2 = 1.5(1 - a^2)$$

this implies $f'(0) > 0, f'(1) = 0, f'(-1) = 0$ we know 1 and -1 are "stable" in the sense that once we reach these two points, model won't go away again, more importantly, $f'(a) > 0$ when $a \in (-1, -1), f'(a) < 0$ when $a \in (-\infty, -1) \cup (1, \infty)$ which implies any point falling in $(-1, \infty)$ will go closer to 1, so the "attractor basin" is any value that, after one step of recursion, do NOT go below -1 [Strange, I lack full proof here] But it can be shown that the "attractor basin" for 1 is when $a_0 \in [0, \sqrt{3}]$ so when we normalize the matrix's singular values to $[0, 1]$, applying this recursion formulation guarantees NS iteration will find $Q = UV^T$. (See below for the recursion illustration)



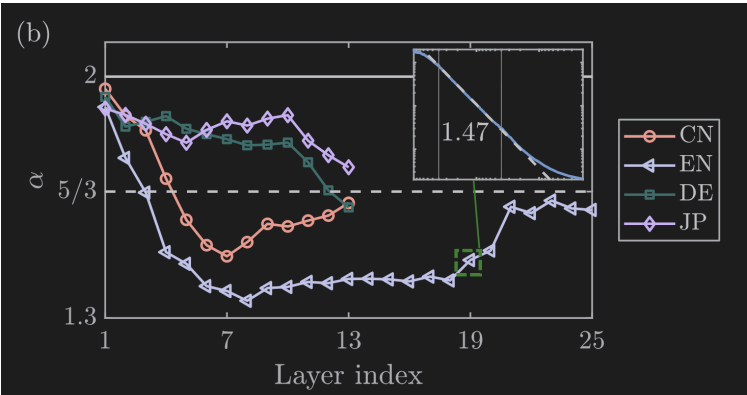
[Connection to SigReg] In SigReg, isotropic Gaussian distributed features are actually preferred as it avoids representation collapse.

[Fourier on representation matrix]

Given representations of a sequence $[r_1, r_2, \dots, r_t]$ [Turbulence-like 5/3 spectral scaling ...] found, if we compute the velocity $v_i = r_i - r_{i-1}$, then a DFT on $v(t)$ would reveals monotonously decreasing signal strength when frequency increases. (For mid layers, speed of decreasing is slower, indicating more "high frequency signals" therein) In another word, low-frequency signal dominates the representation's dynamics.

Puzzling Contradiction.

If I look at attention sink & compression valley & IBLM, I know late layers got smaller representation rank, but in the turbulence paper, we see later layer got less high frequency representations. This implies [Low MBE co-occurs with More high frequency representation] which is honestly strange -- how can more high frequency representation not led to bigger effective rank? I think this is an important intuition to gain.



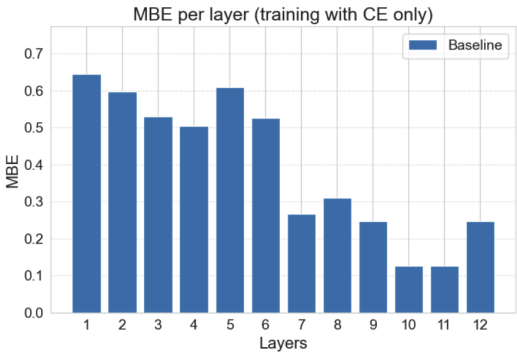
[A plausible explanation] The big BOS token magnitude & attention sink means the "mixed" representations on all token at mid-layer is quite boring: they either attend to (and mix with) BOS token rep (oftenly), or not (rarely), this naturally led to low rank representations since there are so few variation, but it also led to *high frequency* because the "BOS alike representation" simply kept showing up (!) therefore it's a high frequency representation.

[An Idea] We know EN & CN are better performing language, if we are forced to answer "what indicates task success" using the term above (α is the "spectral power scaler" bigger α indicates less high frequency signals), there are few options.

(1). Mid layer (layer 7 in this case) absolute α , smaller the better. We note that CN specifically have higher MBE at the first layer, and DE & CN has very similar end layer & start layer MBE. [An interesting experiment, is then on *What and whether MBE pattern explains task success?* We can try different languages, different tasks etc. Just to understand whether this term is useful and when]

[A hypothesis & idea] Dominating BOS token representation and attention sink reinforce each other, leading to repetitive appearances of "BOS rep like" mixed representations due to the singular attention pattern, this repetition means "frequency" of representations' change are high (it flip back and forth all the time). Therefore, [applying DFT + smooth + DFT which smooth out high frequency filter, might be a nice way of doing regularization] Ideally we can apply this to the representation matrix, along the temporal axis.

(2). Given the inconsistency in different layer's α as a criterion, consider using the "accumulative compression" + "accumulative decompression" as a criterion? This would put CN > JP > DE, EN is not very comparable since it's using GPT model instead of BERT.



[Question] Is there a recursion formulation on $\{a_n\}$ that guarantees "partial convergence to 1 for bigger initial values and to 0 for smaller values, or sth more elegant and beautiful"? What is the most elegant & beautiful form of recursion that can be found also in Nature?
 "bistable recursion"

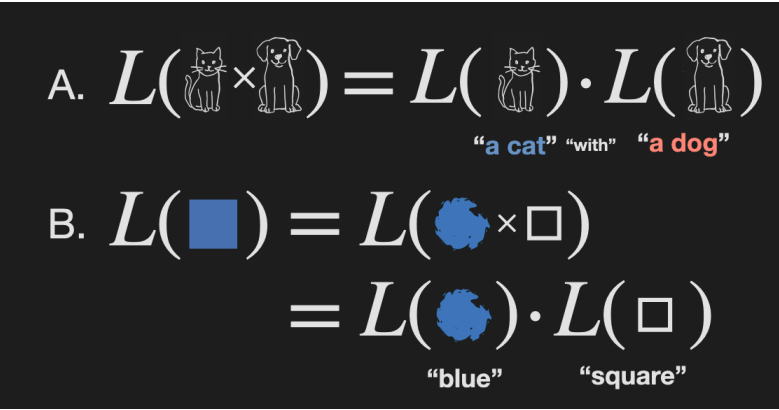
$$a_{n+1} = 3a_n^2 - 2a_n^3$$

initial values in (0, 0.5) converge to 0.0, initial values in (0.5, 1.0) converge to 1.0, while this is nice, if all normalized singular values falls below 0.5, we'd end up with a zero matrix after iteration.

If $(a,b,c) \in [0,1]$ then a very simple way to "spread out" them is by (a^2,b^2,c^2) , which is equivalent to doing $G(G^TG)$.

Compositional generalization means being able to handle novel combinations of familiar concepts --- understanding "pink elephant" even if you've never seen one, just because you know "pink" and "elephant". It's widely believed that [disentangled (compositional structured)] representations are keys to this ability. [So what does it take to ensure disentangled representations?]

Human language is organized in the sense that it has high topological similarity. "Cat as cat, dog as dog" instead of "cat & dog's body as cug", "cat & dog's head as oked". [sequential information processing] offers a perspective:



But the biggest problem with the above diagram, is that it assumes a "meaning space" that accepts "product" operation --- Santa Claus did not offer that to us as free present, without it, this diagram is wishful thinking.

Consider the [end of sentence, or just full sequence?] representation vector from an LLM $r_\theta(\text{cat and dog})$ to be an element in the mysterious & undefined "meaning space". Then the magical property it must satisfies here, is

$$r_\theta(\text{cat}) \odot r_\theta(\text{dog}) = r_\theta(\text{cat and dog})$$

Is there any mathematical framework that resembles this?

Another angle offered, is that language is a structure that minimizes the predictive information or excess entropy E , the mutual information between the past and the future

$$E = I(X_{>t}; X_{\leq t})$$

Applying chain rule of mutual information (conditional on the past, what is the "excessive" mutual information provided by the new token x_i ? this excessive token level entropy reduction accumulates into total mutual information between past, and future sequences)

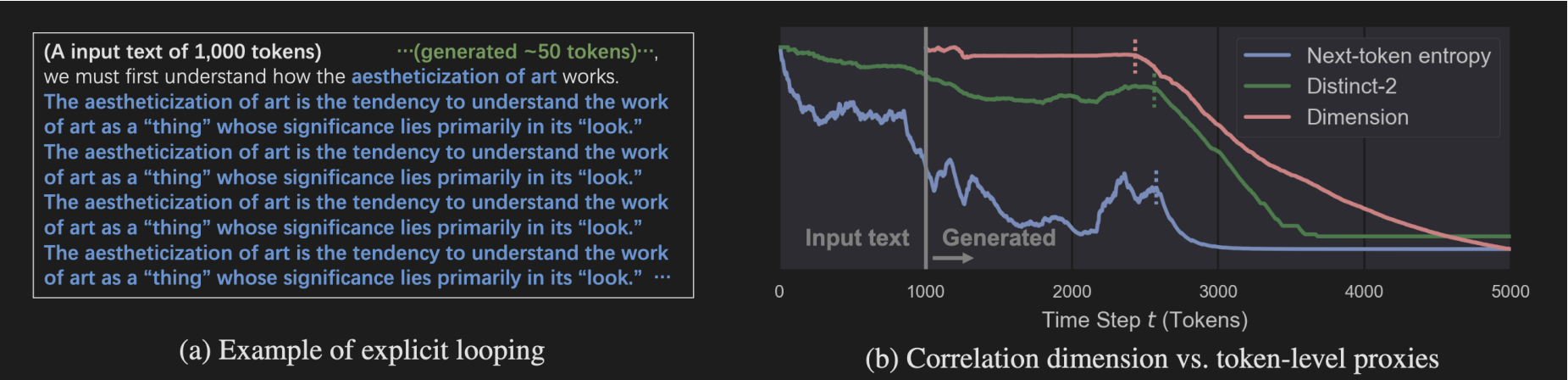
$$\begin{aligned} I(x_{>t}; x_{\leq t}) &= \sum_{i=1}^t I(x_{>t}, x_i | x_{<i}) \\ &= \sum_{i=1}^t H(x_{>t} | x_{<i}) - H(x_{>t} | x_{\leq i}) \end{aligned}$$

each term measures the "information gain" given one new token x_i . BTW, the chain rule in mutual information is

$$I(x; y, z) = I(x; y) + I(x; z | y)$$

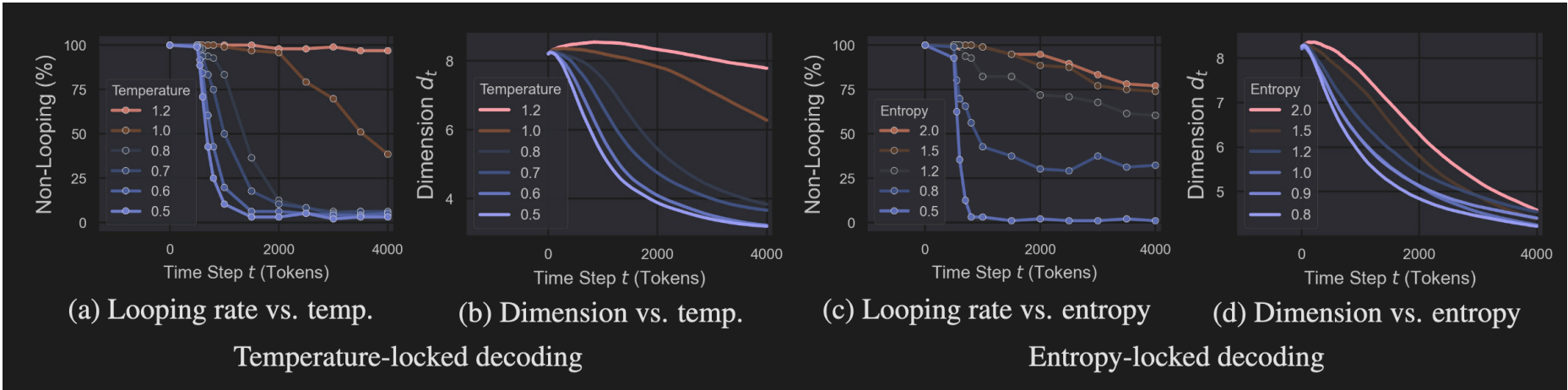
total mutual information between x and (y, z) is the sum of mutual information between (x, y) and the new information about x provided by z conditional on knowing y . Information is additive, which is a nice property.

[Question 1.] Has people study the trend of $H(x_t | x_{<t})$ in LLMs? is it actually monotonously decreasing? this might be a bad news for long horizon agentic performance which involves exploration etc.



"LLM generation via geometric regulation" -- entropy collapse from LLM's rollout which co-occurs with "explicit looping" behavior. Correlated terms

are low temperature sampling scheme, and low *effective* dimension in internal representation matrix.



A test bed:

- (I). compositional datasets (pre-training type)
 - (II). TinyStories, some datasets with long sequence
 - (III). Arithmetic
- To be trained:
- (1). Fractual Seq model
 - (2). Transformer
 - (3). Transformer with above mentioned regularization / reparamerization tricks.
 - (4). TRM-text